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Three-way valve and oxygen concentrator

Abstract

A three-way valve includes a manifold, a valve body, and a switching mechanism. The manifold is provided with a flow path including first, second and third ports, first, second and third passages leading to the first, second and third ports, and a valve chamber that communicates with the first, second, and third passages. The valve body is accommodated in the valve chamber and is displaceable to a first position at which the first port and the third port communicate with each other or a second position at which the second port and the third port communicate with each other. The switching mechanism switches a position of the valve body to the first position or the second position. The manifold is configured by a plurality of manifold members that have a plate shape and are stacked in a plate thickness direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a continuation of International Application No. PCT/JP2023/038636 filed on Oct. 26, 2023, which claims priority to Japanese Patent Application No. 2022-191721, filed on Nov. 30, 2022. The entire disclosures of these applications are incorporated by reference herein.

BACKGROUND

Technical Field

(2) The present disclosure relates to a three-way valve and an oxygen concentrator including the

three-way valve.

Background Art

(3) There is known an oxygen concentrator that generates high concentration oxygen containing oxygen at a concentration higher than oxygen concentration in air and supplies the oxygen to a user. The oxygen concentrator is used, for example, when a patient (user) who has a disease in the lung and the function of the lung is deteriorated performs oxygen therapy.

(4) In an oxygen concentrator in the related art, a three-way valve is used as a control valve that controls a pressurization and exhaust cycle of two adsorption cylinders. As a three-way valve used in an oxygen concentrator, for example, a three-way valve disclosed in Japanese Laid-Open Patent Publication No. 2020-168087 is known. The three-way valve disclosed in Japanese Laid-Open Patent Publication No. 2020-168087 is an internal pilot type three-way valve. The internal pilot type three-way valve includes a valve body including a coupling rod and a diaphragm, a manifold provided with a space that accommodates the valve body and a flow path of air inside, a pilot valve that switches an application destination of a pilot pressure, and the like, and is configured to move the valve body by the pilot pressure to switch the flow path that communicates with the two adsorption cylinders to a pressurization side and an exhaust side. The three-way valve disclosed in Japanese Laid-Open Patent Publication No. 2020-168087 is a double three-way valve having a structure in which two three-way valves are coupled and integrated (hereinafter, also referred to as an internal pilot type double three-way valve).

SUMMARY

(5) A three-way valve of the present disclosure includes a manifold provided with a flow path including a first port, a first passage leading to the first port, a second port, a second passage leading to the second port, a third port, a third passage leading to the third port, and a valve chamber that communicates with the first passage, the second passage, and the third passage, a valve body that is accommodated in the valve chamber and is displaceable to a first position at which the first port and the third port communicate with each other or a second position at which the second port and the third port communicate with each other, and a switching mechanism that switches a position of the valve body to the first position or the second position, in which the manifold is configured by a plurality of manifold members that has a plate shape and is stacked in a plate thickness direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic sectional view of a three-way valve (Embodiment 1) of the present disclosure.

(2) FIG. 2 is a schematic perspective view of the three-way valve (Embodiment 1) of the present disclosure.

(3) FIG. 3A is a top view of a first manifold member in a first three-way valve.

(4) FIG. 3B is a bottom view of the first manifold member in the first three-way valve.

(5) FIG. 3C is a rear view of the first manifold member in the first three-way valve.

(6) FIG. 4A is a top view of a second manifold member in the first three-way valve.

(7) FIG. 4B is a bottom view of the second manifold member in the first three-way valve.

(8) FIG. 5A is a top view of a third manifold member in the first three-way valve.

(9) FIG. 5B is a bottom view of the third manifold member in the first three-way valve.

(10) FIG. 6A is a top view of a fourth manifold member in the first three-way valve.

(11) FIG. 6B is a bottom view of the fourth manifold member in the first three-way valve.

(12) FIG. 7A is an explanatory diagram of an operation (first mode) of the three-way valve of the present disclosure.

- (13) FIG. 7B is an explanatory diagram of an operation (second mode) of the three-way valve of the present disclosure.
- (14) FIG. 8 is a schematic sectional view of a three-way valve (Embodiment 2) of the present disclosure.
- (15) FIG. 9 is a schematic perspective view of the three-way valve (Embodiment 2) of the present disclosure.
- (16) FIG. 10A is a top view of a first manifold member in a second three-way valve.
- (17) FIG. 10B is a bottom view of the first manifold member in the second three-way valve.
- (18) FIG. 10C is a rear view of the first manifold member in the second three-way valve.
- (19) FIG. 11A is a top view of a second manifold member in the second three-way valve.
- (20) FIG. 11B is a bottom view of the second manifold member in the second three-way valve.
- (21) FIG. 12A is a top view of a third manifold member in the second three-way valve.
- (22) FIG. 12B is a bottom view of the third manifold member in the second three-way valve.
- (23) FIG. 13A is a top view of a fourth manifold member in the second three-way valve.
- (24) FIG. 13B is a bottom view of the fourth manifold member in the second three-way valve.
- (25) FIG. 14A is a top view of a fifth manifold member in the second three-way valve.
- (26) FIG. 14B is a bottom view of the fifth manifold member in the second three-way valve.
- (27) FIG. 15 is an explanatory diagram of an oxygen concentrator according to one embodiment of the present disclosure.
- (28) FIG. 16 is a block diagram for describing an oxygen concentration process of the oxygen concentrator.
- (29) FIG. 17 is a diagram for describing a relationship between a pressure change of one cycle of an adsorption cylinder and a switching state of a control valve of the oxygen concentrator.

DETAILED DESCRIPTION OF EMBODIMENT(S)

- (30) Hereinafter, an oxygen supply device of the present disclosure will be described in detail with reference to the accompanying drawings. The present disclosure should not be limited to the following exemplification, but is intended to include any modification recited in the claims within meanings and a scope equivalent to the scope of the claims.
- (31) (Overall Configuration of Three-Way Valve (Embodiment 1) of Present Disclosure)
- (32) FIG. 1 is a schematic sectional view of a three-way valve (Embodiment 1) of the present disclosure. FIG. 2 is a schematic perspective view of the three-way valve of the present disclosure. FIGS. 1 and 2 show a three-way valve 10 as an example of the three-way valve of the present disclosure. The three-way valve 10 shown in FIG. 1 is a three-way valve 10 according to Embodiment 1 of the present disclosure, and is also referred to as first three-way valve 10A in the following description. In the following description, when simply referred to as “three-way valve 10”, a common configuration between the first three-way valve 10A according to Embodiment 1 and a three-way valve 10 according to Embodiment 2 (second three-way valve 10B described later) will be described.
- (33) The three-way valve 10 shown in FIGS. 1 and 2 is an internal pilot type double three-way valve. In the present embodiment, a three-way valve of the present disclosure will be described by exemplifying a three-way valve 10 which is an internal pilot type double three-way valve. In the present embodiment, the three-way valve 10 which is an internal pilot type double three-way valve will be described as an example. However, the three-way valve of the present disclosure may be a single (not double) three-way valve, and is not required to be a three-way valve activated by a pilot pressure.
- (34) As shown in FIGS. 1 and 2, the three-way valve 10 includes a control valve 20, a pilot mechanism 30, and a manifold 50. In the three-way valve 10 according to the present embodiment, the control valve 20 includes a first control valve 21 and a second control valve 22. In the three-way valve 10 according to the present embodiment, the pilot mechanism 30 includes a first pilot valve 31 and a second pilot valve 32. The manifold 50 includes a first valve chamber 51 that

accommodates the first control valve **21** and a second valve chamber **52** that accommodates the second control valve **22**. In such a manner, the three-way valve **10** shown in FIGS. **1** and **2** is a pilot type double three-way valve including a pair of the first control valve **21** and the second control valve **22**, and a pair of the first pilot valve **31** and the second pilot valve **32**. In the present embodiment, the three-way valve of the present disclosure will be described by exemplifying the three-way valve **10** which is a pilot type double three-way valve. In the following description, an X direction, a Y direction, and a Z direction are defined for the three-way valve **10** as follows. The X direction in this description is a direction including an arrangement direction of the first control valve **21** and the second control valve **22** and a direction parallel to the arrangement direction. The Z direction in this description is a direction including a displacement direction of each valve body (a first valve body **23** and a second valve body **24** described later) included in the first control valve **21** and the second control valve **22** and a direction parallel to the displacement direction. The Y direction is a direction perpendicular to the X direction and the Z direction. In the following description, in each drawing, a direction indicated by an arrow in the X direction is also referred to as a right side, an opposite direction to the direction is also referred to as a left side, a direction indicated by an arrow in the Y direction is also referred to as a front side, an opposite direction to the front side is also referred to as a rear side, a direction indicated by an arrow in the Z direction is also referred to as an upper side, and an opposite direction to the upper side is also referred to as a lower side.

(35) (Control Valve)

(36) As shown in FIGS. **1** and **2**, the three-way valve **10** includes the first control valve **21** and the second control valve **22** arranged in a left-right direction (X direction).

(37) The first control valve **21** includes the first valve body **23**. The first valve body **23** includes a first coupling rod **23a**, a first diaphragm **23b** fixed to one end of the first coupling rod **23a**, a second diaphragm **23c** fixed to another end of the first coupling rod **23a**, a first valve portion **23d** provided closer to the first diaphragm **23b** of the first coupling rod **23a**, and a second valve portion **23e** provided closer to the second diaphragm **23c** of the first coupling rod **23a**. The first valve body **23** is accommodated in the first valve chamber **51** of the manifold **50** in such an orientation that an axial direction of the first coupling rod **23a** faces in the Z direction. The first diaphragm **23b** defines a first chamber **A1**, and forms a pilot chamber **A1a** on one side (upper side in the present embodiment) across the first diaphragm **23b**. The first diaphragm **23b** is deformed in the first chamber **A1** in response to a pressure of a pilot fluid supplied to the pilot chamber **A1a**. The first coupling rod **23a** is displaced in the Z direction along with the deformation of the first diaphragm **23b**. The second diaphragm **23c** is deformed in a second chamber **A2** along with the displacement of the first coupling rod **23a**.

(38) The second control valve **22** includes the second valve body **24**. The second valve body **24** includes a second coupling rod **24a**, a third diaphragm **24b** fixed to one end of the second coupling rod **24a**, a fourth diaphragm **24c** fixed to another end of the second coupling rod **24a**, a third valve portion **24d** provided closer to the third diaphragm **24b** of the second coupling rod **24a**, and a fourth valve portion **24e** provided closer to the fourth diaphragm **24c** of the second coupling rod **24a**. The second valve body **24** is accommodated in the second valve chamber **52** of the manifold **50** in such an orientation that an axial direction of the second coupling rod **24a** faces in the Z direction. The third diaphragm **24b** defines a third chamber **A3**, and forms a pilot chamber **A3a** on one side (upper side in the present embodiment) across the third diaphragm **24b**. The third diaphragm **24b** is deformed in the third chamber **A3** in response to a pressure of a pilot fluid supplied to the pilot chamber **A3a**. The second coupling rod **24a** is displaced in the Z direction along with the deformation of the third diaphragm **24b**. The fourth diaphragm **24c** is deformed in the fourth chamber **A4** along with the deformation of the second coupling rod **24a**.

(39) (Pilot Valve)

(40) The first pilot valve **31** is an electromagnetic valve that controls displacement (position

switching) of the first valve body **23**. The second pilot valve **32** is an electromagnetic valve that controls displacement (position switching) of the second valve body **24**. Both of the first pilot valve **31** and the second pilot valve **32** according to the present embodiment are three-port valves. In FIG. **1** (the same applies to FIG. **8** described later), numbers “1”, “2”, or “3” assigned to near the first pilot valve **31** and the second pilot valve **32** represent port numbers of the valves.

(41) (Manifold)

(42) As shown in FIGS. **1** and **2**, in the manifold **50**, the first valve chamber **51** that accommodates the first valve body **23** includes the first chamber **A1** that accommodates the first diaphragm **23b**, the second chamber **A2** that accommodates the second diaphragm **23c**, and a first communication hole **B1** that is a space that allows the first chamber **A1** and the second chamber **A2** to communicate with each other and accommodates the first coupling rod **23a**. The manifold **50** includes, in the first valve chamber **51**, the first valve seat **53** formed at an end of the first communication hole **B1** closer to the first chamber **A1** and a second valve seat **54** formed at an end of the first communication hole **B1** closer to the second chamber **A2**. The first communication hole **B1** is formed to have an axial direction parallel to the Z direction. In the first valve chamber **51**, the first valve body **23** is configured to be displaceable in the Z direction.

(43) The first valve seat **53** faces the first valve portion **23d** of the first diaphragm **23b** accommodated in the first chamber **A1**. In the first control valve **21**, when the first valve portion **23d** and the first valve seat **53** are apart from each other, the fluid can flow between the first chamber **A1** (a portion on another side (lower side) across the first diaphragm **23b**) and the first communication hole **B1**, and when the first valve portion **23d** and the first valve seat **53** are in close contact with each other, the fluid cannot flow between the first chamber **A1** and the first communication hole **B1**. The second valve seat **54** faces the second valve portion **23e** of the second diaphragm **23c** accommodated in the second chamber **A2**. In the first control valve **21**, the fluid can flow between the second chamber **A2** and the first communication hole **B1** when the second valve portion **23e** and the second valve seat **54** are apart from each other, and the fluid cannot flow between the second chamber **A2** and the first communication hole **B1** when the second valve portion **23e** and the second valve seat **54** are in close contact.

(44) Among the positions of the first control valve **21**, a position where the first valve portion **23d** and the first valve seat **53** are in close contact with each other and the second valve portion **23e** and the second valve seat **54** are apart from each other is referred to as a first position **P1**, and a position where the first valve portion **23d** and the first valve seat **53** are apart from each other and the second valve portion **23e** and the second valve seat **54** are in close contact with each other is referred to as a second position **P2**.

(45) In the manifold **50**, the second valve chamber **52** accommodating the second valve body **24** includes the third chamber **A3** accommodating the third diaphragm **24b**, the fourth chamber **A4** accommodating the fourth diaphragm **24c**, and the second communication hole **B2** that is a space that allows the third chamber **A3** the fourth chamber **A4** to communicate with each other and accommodating the second coupling rod **24a**. The manifold **50** includes a third valve seat **55** formed at an end of the second communication hole **B2** closer to the third chamber **A3** and a fourth valve seat **56** formed at an end of the second communication hole **B2** closer to the fourth chamber **A4** in the second valve chamber **52**. The second communication hole **B2** is formed to have an axial direction parallel to the Z direction. In the second valve chamber **52**, the second valve body **24** is configured to be displaceable in the Z direction.

(46) The third valve seat **55** faces the third valve portion **24d** of the third diaphragm **24b** accommodated in the third chamber **A3**. In the second control valve **22**, when the third valve portion **24d** and the third valve seat **55** are apart from each other, the fluid can flow between the third chamber **A3** (a portion on another side (lower side) across the third diaphragm **24b**) and the second communication hole **B2**, and when the third valve portion **24d** and the third valve seat **55** are in close contact with each other, the fluid cannot flow between the third chamber **A3** and the

second communication hole B2. The fourth valve seat 56 faces the fourth valve portion 24e of the fourth diaphragm 24c accommodated in the fourth chamber A4. In the second control valve 22, when the fourth valve portion 24e and the fourth valve seat 56 are apart from each other, the fluid can flow between the fourth chamber A4 and the second communication hole B2, and when the fourth valve portion 24e and the fourth valve seat 56 are in close contact with each other, the fluid cannot flow between the fourth chamber A4 and the second communication hole B2.

(47) Among the positions of the second control valve 22, a position where the third valve portion 24d and the third valve seat 55 are in close contact with each other and the fourth valve portion 24e and the fourth valve seat 56 are apart from each other is referred to as a first position P1, and a position where the third valve portion 24d and the third valve seat 55 are apart from each other and the fourth valve portion 24e and the fourth valve seat 56 are in close contact with each other is referred to as a second position P2. In other words, in the first control valve 21 and the second control valve 22, a position when the first valve body 23 and the second valve body 24 are displaced to an uppermost side is referred to as the second position P2, and a position when the first valve body 23 and the second valve body 24 are displaced to a lowermost side is referred to as the first position P1.

(48) In the manifold 50, a second connecting passage C2 that allows the first chamber A1 and the third chamber A3 to communicate with each other and a first connecting passage C1 that allows the second chamber A2 and the fourth chamber A4 to communicate with each other are formed inside.

(49) The manifold 50 includes an air supply port 71, an exhaust port 72, a first air supply/exhaust port 73, and a second air supply/exhaust port 74 on an outer surface of the manifold 50. The three-way valve 10 of the present disclosure switches the positions of the first valve body 23 of the first control valve 21 and the second valve body 24 of the second control valve 22 to allow one of the first air supply/exhaust port 73 or the second air supply/exhaust port 74 to communicate with the air supply port 71 while allowing another of the first air supply/exhaust port 73 and the second air supply/exhaust port 74 to communicate with the exhaust port 72. In the three-way valve 10 according to the present embodiment, the air supply port 71 and the exhaust port 72 are shared by the first control valve 21 and the second control valve 22. However, the first control valve 21 may have a dedicated air supply port and a dedicated exhaust port, and the second control valve 22 may have a dedicated air supply port and a dedicated exhaust port.

(50) The manifold 50 includes an air supply passage 75 that allows the air supply port 71 and the first connecting passage C1 to communicate with each other, an exhaust passage 76 that allows the exhaust port 72 and the second connecting passage C2 to communicate with each other, a first air supply/exhaust passage 77 that allows the first air supply/exhaust port 73 and the first valve chamber 51 to communicate with each other, and a second air supply/exhaust passage 78 that allows the second air supply/exhaust port 74 and the second valve chamber 52 to communicate with each other.

(51) The manifold 50 includes, on the outer surface, a first pilot port 81, a second pilot port 82, and a third pilot port 83 to which the first pilot valve 31 is connected, and a fourth pilot port 84, a fifth pilot port 85, and a sixth pilot port 86 to which the second pilot valve 32 is connected. The manifold 50 includes a first pilot passage 91 that allows the first pilot port 81 and the first valve chamber 51 to communicate with each other, a second pilot passage 92 that allows the fourth pilot port 84 and the second valve chamber 52 to communicate with each other, a third pilot passage 93 that allows the second pilot port 82 and the fifth pilot port 85 to communicate with the first connecting passage C1, and a fourth pilot passage 94 that allows the third pilot port 83 and the sixth pilot port 86 to communicate with the second connecting passage C2.

(52) As shown in FIG. 1, the first pilot valve 31 and the second pilot valve 32 are three-port valves. The first pilot valve 31 is attached to a rear surface of the manifold 50 such that the first pilot passage 91 is connected to the first port "1", the third pilot passage 93 is connected to the second port "2", and the fourth pilot passage 94 is connected to the third port "3". The second pilot valve

32 is attached to the rear surface of the manifold **50** such that the second pilot passage **92** is connected to the first port “1”, the third pilot passage **93** is connected to the second port “2”, and the fourth pilot passage **94** is connected to the third port “3”.

(53) (Manifold of Three-Way Valve According to Embodiment 1)

(54) FIG. 3A is a top view of a first manifold member in the first three-way valve. FIG. 3B is a bottom view of the first manifold member in the first three-way valve. FIG. 3C is a rear view of the first manifold member in the first three-way valve. FIG. 4A is a top view of a second manifold member in the first three-way valve. FIG. 4B is a bottom view of the second manifold member in the first three-way valve. FIG. 5A is a top view of a third manifold member in the first three-way valve. FIG. 5B is a bottom view of the third manifold member in the first three-way valve. FIG. 6A is a top view of a fourth manifold member in the first three-way valve. FIG. 6B is a bottom view of the fourth manifold member in the first three-way valve. As shown in FIGS. 1 and 2, the manifold **50** constituting the first three-way valve **10A** includes four plate-shaped manifold members **60**. In the following description, the manifold **50** in the first three-way valve **10A** is also referred to as a first manifold **50A**.

(55) The first manifold **50** includes a first manifold member **60** (hereinafter, referred to as a first manifold member **61A**), a second manifold member **60** (hereinafter, referred to as a second manifold member **62A**), a third manifold member **60** (hereinafter, referred to as a third manifold member **63A**), and a fourth manifold member **60** (hereinafter, referred to as a fourth manifold member **64A**).

(56) The first manifold **50A** is configured by stacking the first manifold member **61A**, the second manifold member **62A**, the third manifold member **63A**, and the fourth manifold member **64A** in a plate thickness direction in order from the upper side in the axial direction (Z direction) of the first coupling rod **23a** accommodated in the first valve chamber **51** and the second coupling rod **24a** accommodated in the second valve chamber **52**.

(57) As shown in FIGS. 1, 3A, and 3B, the first manifold member **61A** is provided with a part of the first chamber **A1**, a part of the third chamber **A3**, the first pilot port **81**, the second pilot port **82**, the third pilot port **83**, the fourth pilot port **84**, the fifth pilot port **85**, the sixth pilot port **86**, the first pilot passage **91**, the second pilot passage **92**, a part of the third pilot passage **93**, and a part of the fourth pilot passage **94**.

(58) As shown in FIGS. 1, 4A, and 4B, the second manifold member **62A** is provided with a part of the first chamber **A1**, a part of the third chamber **A3**, a part of the first communication hole **B1**, a part of the second communication hole **B2**, a part of the first connecting passage **C1**, a part of the second connecting passage **C2**, the first valve seat **53**, the third valve seat **55**, a part of the first air supply/exhaust passage **77**, a part of the second air supply/exhaust passage **78**, a part of the third pilot passage **93**, and a part of the fourth pilot passage **94**.

(59) In the first manifold **50A**, the first chamber **A1** and the third chamber **A3** are formed across two manifold members **60**, and include the first manifold member **61A** and the second manifold member **62A**.

(60) As shown in FIGS. 1, 5A, and 5B, the third manifold member **63A** is provided with a part of the second chamber **A2**, a part of the fourth chamber **A4**, a part of the first communication hole **B1**, a part of the second communication hole **B2**, a part of the first connecting passage **C1**, a part of the second connecting passage **C2**, the second valve seat **54**, the fourth valve seat **56**, a part of the air supply passage **75**, a part of the exhaust passage **76**, a part of the first air supply/exhaust passage **77**, a part of the second air supply/exhaust passage **78**, a part of the third pilot passage **93**, and a part of the fourth pilot passage **94**.

(61) In the first manifold **50A**, the first connecting passage **C1** and the second connecting passage **C2** are configured by recesses formed in a boundary surface between the second manifold member **62A** and the third manifold member **63A**. Specifically, the first connecting passage **C1** and the second connecting passage **C2** are configured by a space surrounded by a plane formed on a lower

surface (boundary surface) of the second manifold member **62A** and a recess formed on an upper surface (boundary surface) of the third manifold member **63A**. The first connecting passage **C1** and the second connecting passage **C2** may be configured by a space surrounded by a recess formed in the lower surface (boundary surface) of the second manifold member **62A** and a plane formed in the upper surface (boundary surface) of the third manifold member **63A**, or may be configured by a space surrounded by a recess formed in the lower surface (boundary surface) of the second manifold member **62A** and a recess formed in the upper surface (boundary surface) of the third manifold member **63A**. The “boundary surface” between the adjacent manifold members **60** here includes a mating surface where the manifold members **60** are in contact with each other and a virtual surface obtained by extending the mating surface to a portion (space) where the recess exists.

(62) In other words, in the first manifold **50A**, the first connecting passage **C1** and the second connecting passage **C2** are formed in a range including the boundary surface between the adjacent manifold members **60** (the second manifold member **62A** and the third manifold member **63A**). In the present embodiment, a case where both the first connecting passage **C1** and the second connecting passage **C2** are formed in the range including the boundary surface between the adjacent manifold members **60** is exemplified. However, the first manifold **50A** of the present disclosure may have a configuration in which one of the first connecting passage **C1** or the second connecting passage **C2** is drilled in the manifold member **60**.

(63) For example, in a case where the manifold has one metal lump (block) as in the related art, passages such as the first connecting passage **C1** and the second connecting passage **C2** are drilled in the block by using a drill or the like. Since such a passage formed by a drill or the like has a circular sectional shape, a margin other than the passage has to be large, and as a result, the block is increased in size and weight. On the other hand, as in the manifold **50** of the present disclosure, when the plate-shaped manifold members **60** are stacked in the plate thickness direction, passages such as the first connecting passage **C1** and the second connecting passage **C2** can be formed by recesses provided in the range including the boundary surface between the adjacent manifold members **60**. In the manifold **50** of the present disclosure, the passage extending in a direction orthogonal to a stacking direction of the manifold members **60** can be expanded in the front-rear direction and the left-right direction, and a degree of freedom of the sectional shape of such a passage is higher than in the related art. In the manifold **50** having such a configuration, a margin other than the passages can be suppressed, and as a result, the manifold can be reduced in size and weight.

(64) As shown in FIGS. **1**, **6A**, and **6B**, the fourth manifold member **64A** is provided with a part of the second chamber **A2**, a part of the fourth chamber **A4**, the air supply port **71**, a part of the air supply passage **75**, the exhaust port **72**, a part of the exhaust passage **76**, the first air supply/exhaust port **73**, a part of the first air supply/exhaust passage **77**, the second air supply/exhaust port **74**, and a part of the second air supply/exhaust passage **78**.

(65) In the first manifold **50A**, the second chamber **A2** and the fourth chamber **A4** are formed across two manifold members **60**, and include the third manifold member **63A** and the fourth manifold member **64A**.

(66) In the first manifold **50A**, the first communication hole **B1** and the second communication hole **B2** are formed to have axial directions parallel to the stacking direction of the manifold members **60**. In the first manifold **50A**, the first communication hole **B1** and the second communication hole **B2** are formed across two manifold members **60** (the second manifold member **62A** and the third manifold member **63A**).

(67) (Operation of Three-Way Valve)

(68) FIG. **7B** is an explanatory diagram of an operation of the three-way valve of the present disclosure. As shown in FIG. **7**, the three-way valve **10** of the present disclosure is used in a form in which compressed air is supplied to the air supply port **71** and gas is sucked and exhausted from

the exhaust port 72. The three-way valve 10 having such a configuration includes a pair of adsorption cylinders, and is suitable for use in an oxygen concentrator of a vacuum pressure swing adsorption system (VPSA) that sucks and decompresses one adsorption cylinder while compressed air is supplied to the other adsorption cylinder. Although the three-way valve 10 shown in the present embodiment exemplifies a case where air is sucked and exhausted from the exhaust port 72, the three-way valve 10 of the present disclosure may exhaust air from the exhaust port 72 by opening the exhaust port 72 to atmosphere without sucking air. The three-way valve 10 in this case includes a pair of adsorption cylinders, and is suitable for use in an oxygen concentrator of a pressure swing adsorption system (PSA) in which while compressed air is supplied to one of the adsorption cylinders, the other adsorption cylinder is opened to the atmosphere to be decompressed.

(69) In a first mode, the three-way valve 10 switches the “1” port and the “2” port of the first pilot valve 31 to communicate with each other, and switches the “1” port and the “3” port of the second pilot valve 32 to communicate with each other.

(70) At this time, in the first control valve 21, the pilot chamber A1a communicates with the air supply passage 75 via the first pilot passage 91, the first pilot valve 31, the third pilot passage 93, and the first connecting passage C1. As a result, the compressed air is supplied to the pilot chamber A1a (a positive pilot pressure is applied).

(71) In the first control valve 21, when the compressed air is supplied to the pilot chamber A1a, a pilot pressure is applied to the first diaphragm 23b, and the first coupling rod 23a is thus displaced from the first chamber A1 toward the second chamber A2. At this time, the first valve portion 23d is pressed against the first valve seat 53, and a flow of air from the first communication hole B1 to the first chamber A1 (a portion below the first diaphragm 23b) is sealed. In this description, the position of the first valve body 23 at this time is referred to as a first position P1.

(72) Furthermore, in the first control valve 21, at this time, the second valve portion 23e is apart from the second valve seat 54, and the first connecting passage C1 and the first air supply/exhaust passage 77 thus communicate with each other via the first communication hole B1 and the second chamber A2. As a result, air is supplied from the air supply passage 75 to the first air supply/exhaust passage 77 via the first control valve 21.

(73) At this time, in the second control valve 22, the pilot chamber A3a communicates with the exhaust passage 76 via the second pilot passage 92, the second pilot valve 32, the fourth pilot passage 94, and the second connecting passage C2. As a result, air is sucked and exhausted from the pilot chamber A3a (negative pilot pressure is applied).

(74) In the second control valve 22, when air is sucked and exhausted from the pilot chamber A3a, a negative pilot pressure is applied to the third diaphragm 24b, and the second coupling rod 24a is thus displaced from the fourth chamber A4 toward the third chamber A3. At this time, the fourth valve portion 24e is pressed against the fourth valve seat 56, and a flow of air from the fourth chamber A4 to the second communication hole B2 is sealed.

(75) Furthermore, in the second control valve 22, at this time, the third valve portion 24d is apart from the third valve seat 55, and the second connecting passage C2 and the second air supply/exhaust passage 78 thus communicate with each other via the second communication hole B2 and the third chamber A3 (a portion below the third diaphragm 24b). As a result, gas is exhausted from the exhaust passage 76 to the second air supply/exhaust passage 78 via the second control valve 22. In this description, the position of the second valve body 24 at this time is referred to as a second position P1.

(76) Furthermore, in the second control valve 22, the pilot chamber A3a communicates with the exhaust passage 76 via the second pilot passage 92, the second pilot valve 32, the fourth pilot passage 94, and the second connecting passage C2. In the second control valve 22, air is sucked and exhausted from the pilot chamber A3a, and a negative pilot pressure is thus applied to the second valve body 24 (third diaphragm 24b).

(77) In the second control valve **22**, when air is sucked and exhausted from the pilot chamber **A3a**, a negative pilot pressure is applied to the third diaphragm **24b**, and the second valve body **24** (second coupling rod **24a**) is thus reliably displaced from the first position **P1** to the second position **P2**. Therefore, the second control valve **22** can ensure reliability of the operation.

(78) In such a manner, in the first mode, the three-way valve **10** of the present disclosure switches the flow of the fluid so as to allow a flow of supply air from the air supply port **71** toward the first air supply/exhaust port **73** to exhaust the gas from the first air supply/exhaust port **73**, and to allow a flow of exhaust air from the second air supply/exhaust port **74** toward the exhaust port **72** to suck the gas from the second air supply/exhaust port **74**.

(79) In a second mode, the three-way valve **10** of the present disclosure switches the “1” port and the “3” port of the first pilot valve **31** to communicate with each other, and switches the “1” port and the “2” port of the second pilot valve **32** to communicate with each other.

(80) At this time, in the first control valve **21**, the pilot chamber **A1a** communicates with the exhaust passage **76** via the first pilot passage **91**, the first pilot valve **31**, the fourth pilot passage **94**, and the second connecting passage **C2**. As a result, air is sucked and exhausted from the pilot chamber **A1a** (negative pilot pressure is applied).

(81) In the first control valve **21**, air is sucked and exhausted from the pilot chamber **A1a** of the first chamber **A1**, a negative pilot pressure is applied to the first diaphragm **23b**, and the first coupling rod **23a** is thus displaced from the second chamber **A2** in the Z direction toward the first chamber **A1**. At this time, the second valve portion **23e** is pressed against the second valve seat **54**, and a flow of air from the second chamber **A2** to the first communication hole **B1** is sealed.

(82) Furthermore, in the first control valve **21**, at this time, the first valve portion **23d** is apart from the first valve seat **53**, and the second connecting passage **C2** and the first air supply/exhaust passage **77** thus communicate with each other via the first communication hole **B1** and the first chamber **A1** (a portion below the first diaphragm **23b**). As a result, gas is exhausted from the exhaust passage **76** to the first air supply/exhaust passage **77** via the first control valve **21**. In this description, the position of the first valve body **23** at this time is referred to as a second position **P1**.

(83) Furthermore, in the first control valve **21**, the pilot chamber **A1a** communicates with the exhaust passage **76** via the first pilot passage **91**, the first pilot valve **31**, the fourth pilot passage **94**, and the second connecting passage **C2**. In the first control valve **21**, air is sucked and exhausted from the pilot chamber **A1a**, and a negative pilot pressure is thus applied to the first valve body **23** (first diaphragm **23b**).

(84) In the first control valve **21**, when air is sucked and exhausted from the pilot chamber **A1a**, a negative pilot pressure is applied to the first diaphragm **23b**, and the first valve body **23** (first coupling rod **23a**) is thus reliably displaced from the first position **P1** to the second position **P2**. Therefore, the first control valve **21** can ensure the reliability of the operation.

(85) At this time, in the second control valve **22**, the pilot chamber **A3a** communicates with the air supply passage **75** via the second pilot passage **92**, the second pilot valve **32**, the third pilot passage **93**, and the first connecting passage **C1**. As a result, the compressed air is supplied to the pilot chamber **A3a** (a positive pilot pressure is applied).

(86) In the second control valve **22**, when the compressed air is supplied to the pilot chamber **A3a**, a positive pilot pressure is applied to the third diaphragm **24b**, and the second coupling rod **24a** is thus displaced from the third chamber **A3** toward the fourth chamber **A4** in the Z direction. At this time, the third valve portion **24d** is pressed against the third valve seat **55**, and a flow of air from the third chamber **A3** (a portion below the third diaphragm **24b**) to the second communication hole **B2** is sealed. In this description, the position of the second valve body **24** at this time is referred to as a first position **P1**.

(87) Furthermore, in the second control valve **22**, at this time, the fourth valve portion **24e** is apart from the fourth valve seat **56**, and the first connecting passage **C1** and the second air supply/exhaust passage **78** thus communicate with each other via the second communication hole

B2 and the fourth chamber A4. As a result, air is supplied from the air supply passage 75 to the second air supply/exhaust passage 78 via the second control valve 22.

(88) In such a manner, in the second mode, the three-way valve 10 of the present disclosure switches the flow of the fluid so as to allow a flow of the supply air from the air supply port 71 toward the second air supply/exhaust port 74 to exhaust the gas from the second air supply/exhaust port 74, and to allow a flow of the exhaust air from the first air supply/exhaust port 73 toward the exhaust port 72 to suck the gas from the first air supply/exhaust port 73.

(89) In the three-way valve 10 of the present disclosure, by alternately switching between the first mode and the second mode, the first air supply/exhaust port 73 can be alternately used as an air supply port and an exhaust port, and the second air supply/exhaust port 74 can be alternately used as a port on a side different from the first air supply/exhaust port out of the air supply port and the exhaust port.

(90) {Three-Way Valve According to Embodiment 2}

(91) FIG. 8 is a schematic sectional view of a three-way valve (Embodiment 2) of the present disclosure. FIG. 9 is a schematic perspective view of the three-way valve (Embodiment 2) of the present disclosure. FIGS. 8 and 9 show the second three-way valve 10B that is a three-way valve 10 according to Embodiment 2 of the present disclosure. As shown in FIGS. 8 and 9, in the second three-way valve 10B, the manifold 50 includes five plate-shaped manifold members 60, and the second three-way valve 10B is different from the first three-way valve 10A described above in this respect. In the following description, the manifold 50 in the second three-way valve 10B is also referred to as a second manifold 50B.

(92) (Manifold of Three-Way Valve According to Embodiment 2)

(93) FIG. 10A is a top view of a first manifold member in the second three-way valve. FIG. 10B is a bottom view of the first manifold member in the second three-way valve. FIG. 10C is a rear view of the first manifold member in the second three-way valve. FIG. 11A is a top view of a second manifold member in the second three-way valve. FIG. 11B is a bottom view of the second manifold member in the second three-way valve. FIG. 12A is a top view of a third manifold member in the second three-way valve. FIG. 12B is a bottom view of the third manifold member in the second three-way valve. FIG. 13A is a top view of a fourth manifold member in the second three-way valve. FIG. 13B is a bottom view of the fourth manifold member in the second three-way valve. FIG. 14A is a top view of a fifth manifold member in the second three-way valve. FIG. 14B is a bottom view of the fifth manifold member in the second three-way valve. As shown in FIGS. 8 and 9, the second manifold 50B includes a first manifold member 60 (hereinafter, referred to as a first manifold member 61B), a second manifold member 60 (hereinafter, referred to as a second manifold member 62B), a third manifold member 60 (hereinafter, referred to as a third manifold member 63B), a fourth manifold member 60 (hereinafter, referred to as a fourth manifold member 64B), and a fifth manifold member 60 (hereinafter, referred to as a fifth manifold member 65).

(94) The second manifold 50B is configured by stacking the first manifold member 61B, the second manifold member 62B, the third manifold member 63B, the fourth manifold member 64B, and the fifth manifold member 65 in the plate thickness direction in order from the upper side in the axial direction (Z direction) of the first coupling rod 23a accommodated in the first valve chamber 51 and the second coupling rod 24a accommodated in the second valve chamber 52.

(95) As shown in FIGS. 8, 10A, and 10B, the first manifold member 61B is provided with a part of the first chamber A1, a part of the third chamber A3, the first pilot port 81, the second pilot port 82, the third pilot port 83, the fourth pilot port 84, the fifth pilot port 85, the sixth pilot port 86, the first pilot passage 91, the second pilot passage 92, a part of the third pilot passage 93, and a part of the fourth pilot passage 94.

(96) As shown in FIGS. 8, 11A, and 11B, the second manifold member 62B is provided with a part of the first chamber A1, a part of the third chamber A3, a part of the first communication hole B1, a part of the second communication hole B2, a part of the second connecting passage C2, the first

valve seat **53**, the third valve seat **55**, a part of the third pilot passage **93**, and a part of the fourth pilot passage **94**.

(97) In the second manifold **50B**, the first chamber **A1** and the third chamber **A3** are formed across the two manifold members **60**, and are include first manifold member **61B** and the second manifold member **62B**.

(98) As shown in FIGS. **8**, **12A**, and **12B**, the third manifold member **63B** is provided with a part of the first communication hole **B1**, a part of the second communication hole **B2**, a part of the first connecting passage **C1**, a part of the second connecting passage **C2**, a part of the exhaust passage **76**, a part of the first air supply/exhaust passage **77**, a part of the second air supply/exhaust passage **78**, a part of the third pilot passage **93**, and a part of the fourth pilot passage **94**.

(99) In the second manifold **50B**, the second connecting passage **C2** is configured by a recess formed in a range including a boundary surface between the second manifold member **62B** and the third manifold member **63B**. Specifically, the second connecting passage **C2** is configured by a space surrounded by a lower surface (plane) of the second manifold member **62B** and a recess formed on an upper surface of the third manifold member **63B**. The second connecting passage **C2** may be configured by a space surrounded by a recess formed in the lower surface of the second manifold member **62B** and the upper surface (plane) of the third manifold member **63B**, or may be configured by a space surrounded by a recess formed in the lower surface of the second manifold member **62B** and the recess formed in the upper surface of the third manifold member **63B**.

(100) In other words, in the second manifold **50B**, the second connecting passage **C2** is formed in a range including the boundary surface between the adjacent manifold members **60** (the second manifold member **62B** and the third manifold member **63B**).

(101) As shown in FIGS. **8**, **13 A**, and **B**, the fourth manifold member **64B** is provided with a part of the second chamber **A2**, a part of the fourth chamber **A4**, a part of the first communication hole **B1**, a part of the second communication hole **B2**, a part of the first connecting passage **C1**, the second valve seat **54**, the fourth valve seat **56**, a part of the air supply passage **75**, a part of the exhaust passage **76**, a part of the first air supply/exhaust passage **77**, and a part of the second air supply/exhaust passage **78**.

(102) In the second manifold **50B**, the first connecting passage **C1** is configured by a recess formed in a range including a boundary surface between the third manifold member **63B** and the fourth manifold member **64B**. Specifically, the first connecting passage **C1** is configured by a space surrounded by a lower surface (plane) of the third manifold member **63B** and a recess formed on an upper surface of the fourth manifold member **64B**. The first connecting passage **C1** may be configured by a space surrounded by the recess formed in the lower surface of the third manifold member **63B** and the upper surface (plane) of the fourth manifold member **64B**, or may be configured by a space surrounded by the recess formed in the lower surface of the third manifold member **63B** and the recess formed in the upper surface of the fourth manifold member **64B**.

(103) In other words, in the second manifold **50B**, the first connecting passage **C1** is formed in a range including the boundary surface between the adjacent manifold members **60** (the third manifold member **63B** and the fourth manifold member **64B**). In the present embodiment, a case where both the first connecting passage **C1** and the second connecting passage **C2** are formed in the range including the boundary surface between the adjacent manifold members **60** is exemplified. However, the second manifold **50B** of the present disclosure may have a configuration in which one of the first connecting passage **C1** or the second connecting passage **C2** is drilled in the manifold member **60**.

(104) As shown in FIGS. **8**, **14A**, and **14B**, the fifth manifold member **65** is provided with a part of the second chamber **A2**, a part of the fourth chamber **A4**, the air supply port **71**, a part of the air supply passage **75**, the exhaust port **72**, a part of the exhaust passage **76**, the first air supply/exhaust port **73**, a part of the first air supply/exhaust passage **77**, the second air supply/exhaust port **74**, and a part of the second air supply/exhaust passage **78**.

(105) In the second manifold **50B**, the second chamber **A2** and the fourth chamber **A4** are formed across the two manifold members **60**, and include the fourth manifold member **64B** and the fifth manifold member **65**.

(106) In the second manifold **50B**, the first communication hole **B1** and the second communication hole **B2** are formed to have axial directions parallel to the stacking direction of the manifold members **60**. In the second manifold **50B**, the first communication hole **B1** and the second communication hole **B2** are formed across three manifold members **60** (the second manifold member **62B**, the third manifold member **63B**, and the fourth manifold member **64B**).

(107) For example, in a case where two systems of passages (the first connecting passage **C1** and the second connecting passage **C2**) extending in a direction orthogonal to the stacking direction are provided in one third manifold member **63A** as in the first manifold **50A** having a four-layer structure, there is a concern that expansion of each passage in the front-rear direction is restricted, and as a result, it becomes difficult to secure a cross-sectional area of each passage. On the other hand, in a case where one system of passage extending in a direction orthogonal to the stacking direction is provided in each of the third manifold member **63B** and the fourth manifold member **64B** as in the second manifold **50** having a five-layer structure, the restriction on expansion of each passage in the front-rear direction is relaxed, and as a result, the cross-sectional area of each passage (the first connecting passage **C1** and the second connecting passage **C2**) is easily secured.

(108) (Oxygen Concentrator of Present Disclosure)

(109) FIG. **15** is an explanatory diagram of an oxygen concentrator according to one embodiment of the present disclosure. FIG. **16** is a block diagram for describing an oxygen concentration process of the oxygen concentrator. In FIG. **16**, in order to facilitate understanding, some of the components or elements shown in FIG. **15** are simplified or omitted. An oxygen concentrator of the present disclosure is an apparatus that generates high concentration oxygen containing oxygen having an oxygen concentration higher than an oxygen concentration in air and supplies the high concentration oxygen to a user. The oxygen concentrator is used, for example, in home oxygen therapy for providing high concentration oxygen to a respiratory disease patient or the like who is a user.

(110) (Configuration of Oxygen Concentrator M)

(111) As shown in FIGS. **15** and **16**, an oxygen concentrator **M** includes a first adsorption cylinder **101** and a second adsorption cylinder **102**, a compressor **103** that supplies pressurized air to the first adsorption cylinder **101** and the second adsorption cylinder **102**, and an oxygen tank **105** that stores high concentration oxygen. The compressor **103** according to the present embodiment is a pressurization/vacuum combined compressor capable of pressurizing and sucking gas such as air. The compressor **103** supplies pressurized air to the first adsorption cylinder **101** and the second adsorption cylinder **102**, and desorbs and exhausts adsorbed nitrogen-rich gas by decompression. In the present embodiment, the pressurization/vacuum combined compressor **103** is used. However, the oxygen concentrator of the present disclosure may have a configuration in which a vacuum pump as an intake unit is separately provided in the apparatus, separately from the compressor that supplies the pressurized air.

(112) The compressor **103** is activated and various electromagnetic valves and the like described later are activated by a control unit **140** disposed in the apparatus. The control unit **140** includes a storage unit **140a** that stores a program for activating the oxygen concentrator **M**, and a calculation unit **140b** that transmits an activation signal of an electromagnetic valve or the like. The control unit **140** is connected to a battery **141** serving as a power source in a state of not being connected to a power supply, and a display unit **142** that displays an activation state and the like of the oxygen concentrator **M**.

(113) In the casing **109**, a compressor box **110** accommodating the compressor **103** and an exhaust muffler **114**, a cooling fan **112** for cooling the compressor **103**, an air supply filter **116**, and a fan box **130** accommodating an air supply silencing box **113** are provided. The compressor box **110**,

the air supply silencing box **113**, and the fan box **130** constitute a silencing mechanism in the oxygen concentrator M. In other words, the compressor box **110**, the air supply silencing box **113**, and the fan box **130** can reduce noise generated by each device disposed in the casing **109**.

(114) In the casing **109**, a switching valve **111** that switches a flow of the pressurized air from the compressor **103** to the first adsorption cylinder **101** and the second adsorption cylinder **102** and a flow of an exhaust gas from the first adsorption cylinder **101** and the second adsorption cylinder **102** to the compressor **103** is further provided. The switching valve **111** according to the present embodiment includes a first control valve **111A** that is a three-port valve and a second control valve **111B** that is also a three-port valve.

(115) The oxygen concentrator M of the present disclosure employs the three-way valve **10** described above as the switching valve **111**. The first control valve **111A** corresponds to the first control valve **21** described above, and the second control valve **111B** corresponds to the second control valve **22** described above. In FIG. **16**, the numbers “1”, “2”, or “3” assigned to near the labels indicating the valves represent the port numbers of the valves. The three-port valves are assigned with numbers from “1” to “3”, and two-port valves are assigned with number “1” and “2”. The port “A” in the switching valve **111** in FIG. **15** corresponds to the air supply port **71**, the port “B” corresponds to the exhaust port **72**, the port “C” corresponds to the first air supply/exhaust port **73**, and the port “D” corresponds to the second air supply/exhaust port **74**.

(116) The air supply port **180** provided in the casing **109** is provided with a dustproof filter **115** for collecting dust and the like contained in external air introduced into the apparatus. The external air introduced into the casing **109** through the dustproof filter **115** is sucked into the air supply filter **116** through the opening **131** of the fan box **130**, passes through the air supply silencing box **113**, and is sucked into the compressor **103**. The air supply silencing box **113** is disposed in a flow path of air from the air supply filter **116** to the compressor **103**, and reduces noise caused by air supply and compression of the compressor **103**.

(117) The air (pressurized air) compressed and pressurized by the compressor **103** is supplied to the first adsorption cylinder **101** and the second adsorption cylinder **102** via the first control valve **111A** and the second control valve **111B**. The exhaust gas from the first adsorption cylinder **101** and the second adsorption cylinder **102** is decompressed and sucked by the compressor **103** via the first control valve **111A** and the second control valve **111B**, and is exhausted from the exhaust muffler **114** to the outside from an exhaust port **170** via an opening **117** of the compressor box **110**. The heat of the compressor **103** generated by the operation is sucked into the fan box **130** by the cooling fan **112** via the air supply port **180** of the casing **109** and the opening **131** of the fan box **130**, and is cooled by air blown to the compressor **103** by the cooling fan **112**.

(118) An adsorbent that selectively or preferentially adsorbs nitrogen in the pressurized air supplied from the compressor **103** is accommodated inside the first adsorption cylinder **101** and the second adsorption cylinder **102**. As the adsorbent, for example, zeolite or the like can be used. Details of the process of oxygen concentration using the first adsorption cylinder **101** and the second adsorption cylinder **102** will be described later.

(119) Various valves for controlling a flow rate or a flow of a fluid such as high concentration oxygen, that is, a purge valve **118**, check valves **119** and **120**, and a tuning valve **122** are provided in a flow path (a flow path on an outlet side of high concentration oxygen, and in FIG. **15**, a flow path from lower portions of the first adsorption cylinder **101** and the second adsorption cylinder **102** to a first oxygen outlet **150**) downstream of the first adsorption cylinder **101** and the second adsorption cylinder **102**. A micro pressure sensor **128** for detecting respiration of the user is attached to the tuning valve **122**. The tuning valve **122** is switched to an “open” state or a “closed” state in accordance with a detection result of the micro pressure sensor **128**. The oxygen tank **105** is provided upstream of the tuning valve **122** and downstream of the check valves **119** and **120**. A pressure sensor **123** for detecting a pressure abnormality or the like is provided in a gas flow path between the check valves **119** and **120** and the oxygen tank **105**.

(120) The oxygen concentrator M according to the present embodiment is of an oxygen concentrator of the vacuum pressure swing adsorption system (VPSA) in which while air compressed by the compressor **103** is supplied to one adsorption cylinder, the other adsorption cylinder is sucked by the compressor **103** to be decompressed. However, the oxygen concentrator of the present disclosure is not limited to this apparatus, and may be an oxygen concentrator of the pressure swing adsorption system (PSA) in which while air compressed by a compressor is supplied to one adsorption cylinder, the other adsorption cylinder is opened to the atmosphere to be decompressed.

(121) Each of the first control valve **111A** and the second control valve **111B** is a three-port valve, and switches between a pressurized state in which the pressurized air discharged from the compressor **103** is supplied to the first adsorption cylinder **101** (second adsorption cylinder **102**) and a decompressed state in which the exhaust gas in the first adsorption cylinder **101** (second adsorption cylinder **102**) is exhausted to the outside by suction. When one of the adsorption cylinders is in the pressurized state, the other adsorption cylinder is in the decompressed state.

(122) The check valve **119** is disposed in a gas flow path downstream of the first adsorption cylinder **101**, and the check valve **120** is disposed in a gas flow path downstream of the second adsorption cylinder **102**. The check valves **119** and **120** are configured such that the high concentration oxygen exhausted from the first adsorption cylinder **101** and the second adsorption cylinder **102** flows only toward downstream. The purge valve **118** is disposed in a gas flow path connecting a gas flow path between the first adsorption cylinder **101** and the check valve **119** and a gas flow path between the second adsorption cylinder **102** and the check valve **120**.

(123) The high concentration oxygen from the check valve **119** and the high concentration oxygen from the check valve **120** are alternately supplied to the oxygen tank **105** and stored in the oxygen tank **105**. A bacterial filter **125** for removing foreign substances from the high concentration oxygen and a tuning valve **122** for adjusting the flow rate of the high concentration oxygen from the oxygen tank **105** are disposed downstream of the oxygen tank **105**. The high concentration oxygen having the flow rate adjusted by the tuning valve **122** is sent to the oxygen outlet **150** of the casing **109** via an oxygen sensor **124** for detecting an oxygen concentration abnormality. In the oxygen concentrator M, a cannula joint **126** is provided at the oxygen outlet **150**, and the high concentration oxygen is supplied to a patient via a tube TA and a cannula Ca (see FIG. **16**) connected to the cannula joint **126**.

(124) (Oxygen Concentration Process)

(125) FIG. **17** is a diagram for describing a relationship between a pressure change of one cycle of an adsorption cylinder and a switching state of a control valve of the oxygen concentrator. Here, a process of generating high concentration oxygen in the oxygen concentrator M will be described.

(126) In FIG. **17**, the upper diagram shows an open/close state of the first control valve **111A**, the second control valve **111B**, and the purge valve **118** (see FIG. **16**) in each step in relation to the oxygen concentration process, and the lower diagram shows a pressure change in the first adsorption cylinder **101** and the second adsorption cylinder **102** (see FIG. **16**). In the lower diagram, a thick solid line indicates a pressure change inside the first adsorption cylinder **101**, and a thin solid line indicates a pressure change inside the second adsorption cylinder **102**. In the example shown in FIG. **17**, a pressurization process in the adsorption cylinder is performed in the order of the first adsorption cylinder **101** and the second adsorption cylinder **102**. In FIG. **17**, one cycle of processing of the first adsorption cylinder **101** is performed in a period indicated by "T". The processing of one cycle includes six steps from "T1" to "T6" shown in the upper diagram.

(127) In FIG. **16**, the number assigned to each block indicating the first control valve **111A**, the second control valve **111B**, and the purge valve **118** represents the port number of each valve as described above. The first control valve **111A** and the second control valve **111B**, which are three-port valves, are assigned with three numbers from 1 to 3. The purge valve **118**, which is a two-port valve, is assigned with two numbers from 1 to 2. In the upper diagram of FIG. **17**, for example, "1

to 2" of the first control valve **111A** being "open" indicates that a port indicated by "1" to a port indicated by "2" in the first control valve **111A** are in a communicating state. At this time, in the first control valve **111A**, a port indicated by "2" to a port indicated by "3" are in a non-communicating state.

(128) In the lower diagram of FIG. **17**, the horizontal axis indicates a lapse of time, and in the diagram, time passes from the left side to the right side. In step T1, the purge valve **118** is in the "open" state, and the high-concentration oxygen gas in the second adsorption cylinder **102** is supplied from the second adsorption cylinder **102** to the first adsorption cylinder **101**. In step T1, since the ports "2" to "3" of the first control valve **111A** and the second control valve **111B** are both in the "closed" state, the insides of the first adsorption cylinder **101** and the second adsorption cylinder **102** are not sucked. The suction of the first adsorption cylinder **101** and the suction of the second adsorption cylinder **102** are performed such that the timing is deviated from each other by controlling the opening and closing of the valve.

(129) In subsequent step T2, the ports "1" to "2" of the first control valve **111A** and the ports "2" to "3" of the second control valve **111B** are in the "open" state, and pressurization of the first adsorption cylinder **101** and decompression of the second adsorption cylinder **102** by the compressor **103** are performed. In step T2, the purge valve **118** in the "open" state in step T1 is in the "closed" state. In the first adsorption cylinder **101** in the pressurized state by the supply of the pressurized air, nitrogen contained in the pressurized air is adsorbed by the adsorbent accommodated in the first adsorption cylinder **101**. As a result, the gas in the first adsorption cylinder **101** becomes high concentration oxygen having a higher oxygen concentration than a normal oxygen concentration in the air.

(130) In subsequent step T3, the purge valve **118** is in the "open" state, and the high concentration oxygen in the first adsorption cylinder **101** is supplied into the second adsorption cylinder **102** via the purge valve **118**.

(131) In subsequent step T4, the ports "2" to "3" of the first control valve **111A** and the ports "2" to "3" of the second control valve **111B** are in the "closed" state. In step T4, the supply of high concentration oxygen from the first adsorption cylinder **101** to the second adsorption cylinder **102** in step T3 is continued.

(132) In subsequent step T5, the ports "2" to "3" of the first control valve **111A** and the ports "1" to "2" of the second control valve **111B** are in the "open" state, and pressurization of the second adsorption cylinder **102** and decompression of the first adsorption cylinder **101** by the compressor **103** are performed. In step T4, the purge valve **118** in the "open" state in step T3 is in the "closed" state. In the second adsorption cylinder **102** in the pressurized state by the supply of the pressurized air, nitrogen contained in the pressurized air is adsorbed by the adsorbent accommodated in the second adsorption cylinder **102**. As a result, the gas in the second adsorption cylinder **102** becomes high concentration oxygen having a higher oxygen concentration than the normal oxygen concentration in the air.

(133) In subsequent step T6, the purge valve **118** is in the "open" state, and the high concentration oxygen in the second adsorption cylinder **102** is supplied into the first adsorption cylinder **101** via the purge valve **118**. Thereafter, steps T1 to T6 described above are repeated. The oxygen concentrator M generates high concentration oxygen by repeating steps T1 to T6 in the first adsorption cylinder **101** and the second adsorption cylinder **102**, and stores the generated high concentration oxygen in the oxygen tank **105**.

Functional Effects of Embodiments

(134) (1)

(135) The three-way valve **10** according to the above embodiment includes the manifold **50** provided with a flow path including the air supply port **71**, the air supply passage **75** leading to the air supply port **71**, the exhaust port **72**, the exhaust passage **76** leading to the exhaust port **72**, the air supply/exhaust ports **73** and **74**, the air supply/exhaust passages **77** and **78** leading to the air

supply/exhaust ports **73** and **74**, and the valve chambers **51** and **52** that communicate with the air supply passage **75**, the exhaust passage **76** and the air supply/exhaust passages **77** and **78**, the valve bodies **23** and **24** that are accommodated in the valve chambers **51** and **52** and are displaceable to the first position at which the air supply port **71** communicates with the air supply/exhaust passages **77** and **78** or the second position at which the exhaust port **72** communicates with the air supply/exhaust passages **77** and **78**, and the pilot mechanism **30** that switches the positions of the valve bodies **23** and **24** to the first position or the second position. In the three-way valve **10**, the manifold **50** is configured by the plurality of manifold members **60** that has a plate shape and is stacked in the plate thickness direction.

(136) In this three-way valve **10**, by adopting the manifold **50** having a stacked structure, the sectional shape of the passage constituting each flow path in the manifold **50** can be a shape other than circular, and a space efficiency in the manifold **50** (a ratio of a space to be secured in the manifold **50** to a volume of the manifold **50**) can be increased. As a result, the three-way valve **10** can be reduced in size and weight.

(137) (2)

(138) In the three-way valve **10** according to the above embodiment, the flow path provided in the manifold **50** includes the first valve chamber **51**, the second valve chamber **52**, the first connecting passage **C1** that allows the first valve chamber **51** and the second valve chamber **52** to communicate with each other and is connected to the air supply passage **75**, the second connecting passage **C2** that allows the first valve chamber **51** and the second valve chamber **52** to communicate with each other and is connected to the exhaust passage **76**, the first air supply/exhaust port **73** on a side of the first valve chamber **51**, the first air supply/exhaust passage **77** on a side of the first valve chamber **51**, the second air supply/exhaust port **74** on a side of the second valve chamber **52**, and the second air supply/exhaust passage **78** on a side of the second valve chamber **52**. In the three-way valve **10**, the valve body includes the first valve body **23** and the second valve body **24**, the first valve body **23** is accommodated in the first valve chamber **51**, and the second valve body **24** is accommodated in the second valve chamber **52**.

(139) The three-way valve **10** can be reduced in size and weight in a case of being a double three-way valve.

(140) (3)

(141) In the three-way valve **10** according to the above embodiment, the first connecting passage **C1** and the second connecting passage **C2** are constituted by the recess in the boundary surface between the adjacent manifold members **60** of the plurality of manifold members.

(142) The three-way valve **10** with the manifold **50** in a stacked structure enables the first connecting passage **C1** and the second connecting passage **C2** to be formed at a boundary portion of the manifold member **60**. As a result, the space efficiency in the manifold **50** can be increased, and the manifold **50** can be downsized.

(143) (4)

(144) In the three-way valve **10** according to the above embodiment, the flow path provided in the manifold **50** further includes the first pilot passage **91** that communicates the air supply passage **75** or the exhaust passage **76** with the first valve chamber **51**, and the second pilot passage **92** that communicates the air supply passage **75** or the exhaust passage **76** with the second valve chamber **52**. The first valve body **23** includes the first diaphragm **23b** and the second diaphragm **23c** that are deformable by the pressure of the fluid supplied to the first valve chamber **51** via the first pilot passage **91**. The second valve body **24** includes the third diaphragm **24b** and the fourth diaphragm **24c** that are deformable by the pressure of the fluid supplied to the second valve chamber **52** via the second pilot passage **92**. The pilot mechanism **30** includes the first pilot valve **31** and the second pilot valve **32**. In the pilot mechanism **30**, the first pilot valve **31** and the second pilot valve **32** switch a supply destination of the fluid to one of the first pilot passage **91** or the second pilot passage **92**, and the first diaphragm **23b** and the second diaphragm **23c** are deformed or the third

diaphragm **24b** and the fourth diaphragm **24c** are deformed to position one of the first valve body **23** or the second valve body **24** at the first position **P1** and another one of the first valve body **23** or the second valve body **24** at the second position **P2**.

(145) The three-way valve **10** can be reduced in size and weight in a case of being a pilot type three-way valve activated by a pilot pressure.

(146) (5)

(147) In the three-way valve **10** according to the above embodiment, the flow path provided in the manifold **50** further includes the fourth pilot passage **94** that allows the first pilot passage **91** or the second pilot passage **92** and the exhaust passage **76** to communicate with each other. In the three-way valve **10**, the first pilot valve **31** and the second pilot valve **32** switch a communication destination of the fourth pilot passage **94** to one of the first pilot passage **91** or the second pilot passage **92** to deform the first diaphragm **23b** or the third diaphragm **24b**.

(148) In the three-way valve **10**, in a case where air is sucked and exhausted from the exhaust port **72**, it is possible to increase reliability of displacement operations of the first valve body **23** and the second valve body **24**.

(149) (6)

(150) In the three-way valve **10** according to the above embodiment, the first valve body **23** includes the first coupling rod **23a**, the first diaphragm **23b** coupled to one end of the first coupling rod **23a**, the second diaphragm **23c** coupled to the other end of the first coupling rod **23a**, the first valve portion **23d** provided closer to the first diaphragm **23b** of the first coupling rod **23a**, and the second valve portion **23e** provided closer to the second diaphragm **23c** of the first coupling rod **23a**. The second valve body **24** includes the second coupling rod **24a**, the third diaphragm **24b** coupled to one end of the second coupling rod **24a**, the fourth diaphragm **24c** coupled to another end of the second coupling rod **24a**, the third valve portion **24d** provided closer to the third diaphragm **24b** of the second coupling rod **24a**, and the fourth valve portion **24e** provided closer to the fourth diaphragm **24c** of the second coupling rod **24a**. The first valve chamber **51** includes the first chamber **A1** that accommodates the first diaphragm **23b**, the second chamber **A2** that accommodates the second diaphragm **23c**, the first communication hole **B1** that allows the first chamber **A1** and the second chamber **A2** to communicate with each other and accommodates the first coupling rod **23a**, the first valve seat **53** formed at an end of the first communication hole **B1** closer to the first chamber **A1** and facing the first valve portion **23d**, and the second valve seat **54** formed at an end of the first communication hole **B1** closer to the second chamber **A2** and facing the second valve portion **23e**. The second valve chamber **52** includes the third chamber **A3** that accommodates the third diaphragm **24b**, the fourth chamber **A4** that accommodates the fourth diaphragm **24c**, the second communication hole **B2** that allows the third chamber **A3** and the fourth chamber **A4** to communicate with each other and accommodates the second coupling rod **24a**, the third valve seat **55** formed at an end of the second communication hole **B2** closer to the third chamber **A3** and facing the third valve portion **24d**, and the fourth valve seat **56** formed at an end of the second communication hole **B2** closer to the fourth chamber **A4** and facing the fourth valve portion **24e**. The first connecting passage **C1** allows the second chamber **A2** and the fourth chamber **A4** to communicate with each other, the second connecting passage **C2** allows the first chamber **A1** and the third chamber **A3** to communicate with each other, the third pilot passage **93** communicates with the air supply passage **75** via the first access passage **C1**, and the fourth pilot passage **94** communicates with the exhaust passage **76** via the second access passage **C2**.

(151) The three-way valve **10** can be reduced in size and weight in a case of being an internal pilot type double three-way valve.

(152) (7)

(153) In the three-way valve **10** according to the above embodiment, in the manifold **50**, the first communication hole **B1** and the second communication hole **B2** has an axial direction that is parallel to the stacking direction (**Z** direction) of the plurality of manifold members **60**. In the

manifold **50**, the first chamber **A1**, the second chamber **A2**, the third chamber **A3**, the fourth chamber **A4**, the first communication hole **B1**, and the second communication hole **B2** are formed across two or more manifold members **60**.

(154) In the three-way valve **10**, the internal pilot type double three-way valve can be configured by using the manifold **50** having a stacked structure. As a result, the internal pilot type double three-way valve can be reduced in size and weight.

(155) (8)

(156) The oxygen concentrator **M** according to the above embodiment generates high concentration oxygen containing oxygen having a higher concentration than the oxygen concentration in the air and supplies the generated high concentration oxygen. The oxygen concentrator **M** includes an adsorbent that allows adsorption of nitrogen or oxygen contained in the air and desorption of the nitrogen or oxygen adsorbed, the first adsorption cylinder **101** and the second adsorption cylinder **102** that accommodate the adsorbent, the air supply pipe **106** that supplies the air as a raw material of the high concentration oxygen to the first adsorption cylinder **101** and the second adsorption cylinder **102**, the exhaust pipe **107** that exhausts, from the first adsorption cylinder **101** and the second adsorption cylinder **102**, the high concentration oxygen generated, and the switching valve **111** that alternately selects the first adsorption cylinder **101** or the second adsorption cylinder **102**, and connects one of the first adsorption cylinder **101** or the second adsorption cylinder **102** selected to the air supply pipe **106**, while connecting another one of the first adsorption cylinder **101** or the second adsorption cylinder **102** to the exhaust pipe **107**, in which the switching valve **111** includes the three-way valve **10**.

(157) The oxygen concentrator **M**, which employs the three-way valve **10** as the switching valve **111**, can be reduced in size and weight. As the oxygen concentrator **M**, there are a portable type suitable for outdoor use and a stationary type suitable for indoor use. In the oxygen concentrator **M**, it is preferable to employ the three-way valve **10** as the switching valve **111** regardless of the portable type or the stationary type, but in the case of the portable type, it is particularly preferable to employ the three-way valve **10** as the switching valve **111**.

(158) Although the oxygen concentration apparatus of the present disclosure uses a nitrogen adsorbent such as zeolite as an adsorbent, the present disclosure may be applied to an oxygen concentrator using an oxygen adsorbent as an adsorbent.

Claims

1. A three-way valve comprising: a manifold provided with a flow path including a first port, a first passage leading to the first port, a second port, a second passage leading to the second port, a third port, a third passage leading to the third port, and a valve chamber that communicates with the first passage, the second passage, and the third passage; a valve body that is accommodated in the valve chamber and is displaceable to a first position at which the first port and the third port communicate with each other or a second position at which the second port and the third port communicate with each other; and a switching mechanism configured to switch a position of the valve body to the first position or the second position, the manifold being configured by a plurality of manifold members that have a plate shape and are stacked in a plate thickness direction.

2. The three-way valve according to claim 1, wherein the flow path provided in the manifold includes a first valve chamber as a first of the valve chamber, a second valve chamber as a second of the valve chamber, a first connecting passage that allows the first valve chamber and the second valve chamber to communicate with each other and is connected to the first passage, a second connecting passage that allows the first valve chamber and the second valve chamber to communicate with each other and is connected to the second passage, a first valve chamber side third port that is the third port on a side of the first valve chamber, a first valve chamber side third passage that is the third passage on a side of the first valve chamber, a second valve chamber side

third port that is the third port on a side of the second valve chamber, and a second valve chamber side third passage that is the third passage on a side of the second valve chamber, the valve body includes a first valve body as a first of the valve body, and a second valve body as a second of the valve body, and the first valve body is accommodated in the first valve chamber, and the second valve body is accommodated in the second valve chamber.

3. The three-way valve according to claim 2, wherein at least one of the first connecting passage and the second connecting passage is formed by a recess in a boundary surface between adjacent manifold members of the plurality of manifold members.

4. The three-way valve according to claim 2, wherein the flow path provided in the manifold further includes a first pilot passage that communicates with the first valve chamber, a second pilot passage that communicates with the second valve chamber, and a third pilot passage that allows the first pilot passage or the second pilot passage and the first passage to communicate with each other, the first valve body further includes a first diaphragm and a second diaphragm that are deformable by a pressure of a fluid supplied to the first valve chamber via the first pilot passage, the second valve body further includes a third diaphragm and a fourth diaphragm that are deformable by the pressure of the fluid supplied to the second valve chamber via the second pilot passage, the switching mechanism includes a pilot valve that switches a communication destination of the third pilot passage to one of the first pilot passage or the second pilot passage, and the first diaphragm and the second diaphragm are deformed and the third diaphragm and the fourth diaphragm are deformed to position one of the first valve body or the second valve body at the first position and position another one of the first valve body or the second valve body at the second position.

5. The three-way valve according to claim 4, wherein the flow path provided in the manifold further includes a fourth pilot passage that allows the first pilot passage or the second pilot passage and the second passage to communicate with each other, and the pilot valve switches a communication destination of the fourth pilot passage to one of the first pilot passage or the second pilot passage to deform the first diaphragm and the second diaphragm or the third diaphragm and the fourth diaphragm.

6. The three-way valve according to claim 5, wherein the first valve body includes a first coupling rod, the first diaphragm coupled to one end of the first coupling rod, the second diaphragm coupled to another end of the first coupling rod, a first valve portion provided on the first diaphragm, and a second valve portion provided on the second diaphragm, the second valve body includes a second coupling rod, the third diaphragm coupled to one end of the second coupling rod, the fourth diaphragm coupled to another end of the second coupling rod, a third valve portion provided on the third diaphragm, and a fourth valve portion provided on the fourth diaphragm, the first valve chamber includes a first chamber that accommodates the first diaphragm, a second chamber that accommodates the second diaphragm, a first communication hole that allows the first chamber and the second chamber to communicate with each other and accommodates the first coupling rod, a first valve seat formed at an end of the first communication hole closer to the first chamber and facing the first valve portion, and a second valve seat formed at an end of the first communication hole closer to the second chamber and facing the second valve portion, the second valve chamber includes a third chamber that accommodates the third diaphragm, a fourth chamber that accommodates the fourth diaphragm, a second communication hole that allows the third chamber and the fourth chamber to communicate with each other and accommodates the second coupling rod, a third valve seat formed at an end of the second communication hole closer to the third chamber and facing the third valve portion, and a fourth valve seat formed at an end of the second communication hole closer to the fourth chamber and facing the fourth valve portion, the first connecting passage allows the second chamber and the fourth chamber to communicate with each other, the second connecting passage allows the first chamber and the third chamber to communicate with each other, the third pilot passage communicates with the first passage via the first connecting passage, and the fourth pilot passage communicates with the second passage via

the second connecting passage.

7. The three-way valve according to claim 6, wherein in the manifold, each of the first communication hole and the second communication hole has an axial direction that is parallel to a stacking direction of the plurality of manifold members, and the first chamber, the second chamber, the third chamber, the fourth chamber, the first communication hole, and the second communication hole are formed across two or more manifold members of the plurality of manifold members.

8. An oxygen concentrator including the three-way valve according to claim 4, the oxygen concentrator generating high concentration oxygen containing oxygen at a concentration higher than an oxygen concentration in air and supplies the high concentration oxygen generated, the oxygen concentrator further comprising: an adsorbent that allows adsorption of nitrogen or oxygen contained in the air and desorption of the nitrogen or oxygen adsorbed; a first adsorption cylinder and a second adsorption cylinder that accommodate the adsorbent; an air supply pipe that supplies the air as a raw material of the high concentration oxygen to the first adsorption cylinder and the second adsorption cylinder; an exhaust pipe that exhausts, from the first adsorption cylinder and the second adsorption cylinder, the high concentration oxygen generated; and a switching valve that alternately selects the first adsorption cylinder or the second adsorption cylinder, and connects one of the first adsorption cylinder or the second adsorption cylinder selected to the air supply pipe while connecting another one of the first adsorption cylinder or the second adsorption cylinder to the exhaust pipe, the switching valve including the three-way valve.

9. The three-way valve according to claim 3, wherein the flow path provided in the manifold further includes a first pilot passage that communicates with the first valve chamber, a second pilot passage that communicates with the second valve chamber, and a third pilot passage that allows the first pilot passage or the second pilot passage and the first passage to communicate with each other, the first valve body further includes a first diaphragm and a second diaphragm that are deformable by a pressure of a fluid supplied to the first valve chamber via the first pilot passage, the second valve body further includes a third diaphragm and a fourth diaphragm that are deformable by the pressure of the fluid supplied to the second valve chamber via the second pilot passage, the switching mechanism includes a pilot valve that switches a communication destination of the third pilot passage to one of the first pilot passage or the second pilot passage, and the first diaphragm and the second diaphragm are deformed and the third diaphragm and the fourth diaphragm are deformed to position one of the first valve body or the second valve body at the first position and position another one of the first valve body or the second valve body at the second position.

10. The three-way valve according to claim 9, wherein the flow path provided in the manifold further includes a fourth pilot passage that allows the first pilot passage or the second pilot passage and the second passage to communicate with each other, and the pilot valve switches a communication destination of the fourth pilot passage to one of the first pilot passage or the second pilot passage to deform the first diaphragm and the second diaphragm or the third diaphragm and the fourth diaphragm.

11. The three-way valve according to claim 10, wherein the first valve body includes a first coupling rod, the first diaphragm coupled to one end of the first coupling rod, the second diaphragm coupled to another end of the first coupling rod, a first valve portion provided on the first diaphragm, and a second valve portion provided on the second diaphragm, the second valve body includes a second coupling rod, the third diaphragm coupled to one end of the second coupling rod, the fourth diaphragm coupled to another end of the second coupling rod, a third valve portion provided on the third diaphragm, and a fourth valve portion provided on the fourth diaphragm, the first valve chamber includes a first chamber that accommodates the first diaphragm, a second chamber that accommodates the second diaphragm, a first communication hole that allows the first chamber and the second chamber to communicate with each other and accommodates the first coupling rod, a first valve seat formed at an end of the first communication

hole closer to the first chamber and facing the first valve portion, and a second valve seat formed at an end of the first communication hole closer to the second chamber and facing the second valve portion, the second valve chamber includes a third chamber that accommodates the third diaphragm, a fourth chamber that accommodates the fourth diaphragm, a second communication hole that allows the third chamber and the fourth chamber to communicate with each other and accommodates the second coupling rod, a third valve seat formed at an end of the second communication hole closer to the third chamber and facing the third valve portion, and a fourth valve seat formed at an end of the second communication hole closer to the fourth chamber and facing the fourth valve portion, the first connecting passage allows the second chamber and the fourth chamber to communicate with each other, the second connecting passage allows the first chamber and the third chamber to communicate with each other, the third pilot passage communicates with the first passage via the first connecting passage, and the fourth pilot passage communicates with the second passage via the second connecting passage.

12. The three-way valve according to claim 11, wherein in the manifold, each of the first communication hole and the second communication hole has an axial direction that is parallel to a stacking direction of the plurality of manifold members, and the first chamber, the second chamber, the third chamber, the fourth chamber, the first communication hole, and the second communication hole are formed across two or more manifold members of the plurality of manifold members.

13. An oxygen concentrator including the three-way valve according to claim 9, the oxygen concentrator generating high concentration oxygen containing oxygen at a concentration higher than an oxygen concentration in air and supplies the high concentration oxygen generated, the oxygen concentrator further comprising: an adsorbent that allows adsorption of nitrogen or oxygen contained in the air and desorption of the nitrogen or oxygen adsorbed; a first adsorption cylinder and a second adsorption cylinder that accommodate the adsorbent; an air supply pipe that supplies the air as a raw material of the high concentration oxygen to the first adsorption cylinder and the second adsorption cylinder; an exhaust pipe that exhausts, from the first adsorption cylinder and the second adsorption cylinder, the high concentration oxygen generated; and a switching valve that alternately selects the first adsorption cylinder or the second adsorption cylinder, and connects one of the first adsorption cylinder or the second adsorption cylinder selected to the air supply pipe while connecting another one of the first adsorption cylinder or the second adsorption cylinder to the exhaust pipe, the switching valve including the three-way valve.
